**Even End Semester Examination 2025**

*Name of the Program*: **B. Tech Biotechnology** *Semester:* **4th**
*Name of the Course*: **Waste Management & Upcycling** *Course Code:* **TBT 407**

Time: **3hrs** Maximum Marks: **100**

**Note:**

i. Answer **all the questions** by choosing ***any two of the sub questions.***
ii. Each question carries 20 marks

| | | | |
|---|---|---|---|
| **Q1** | (20 marks) | | **CO1** |
| (a) | Explain the global scenario of waste generation | | |
| | **OR** | | |
| (b) | Match the circular economy principle with its corresponding real-world example: | | |
| | **Column A** | **Column B** | |
| | A. Design out waste | 1. Modular smartphones with replaceable components | |
| | B. Keep products/materials in use | 2. Upcycling furniture from discarded wo pallets | |
| | C. Regenerate natural systems | 3. Composting organic waste for soil enrichment | |
| | D. Extend product lifecycle | 4. Repair cafés or electronic repair works | |
| | E. Industrial symbiosis | 5. Using fly ash from power plants in cem industry | |
| | **OR** | | |
| (c) | Explain how waste upcycling can help in addressing the challenge of waste management? | | |
| **Q2** | (20 marks) | | **CO2** |
| (a) | Discuss the factors influencing waste generation and its health impacts. | | |
| | **OR** | | |
| (b) | Explain how multi-criteria analysis assists in waste management. | | |
| | **OR** | | |
| (c) | What is lifecycle analysis and how is it used in waste management? | | |
| **Q3** | (20 marks) | | **CO3** |
| (a) | Analyze the impact of improper landfill management on environmental pollution? Implement strategies for effective landfill management? | | |
| | **OR** | | |
| (b) | Assess decision support tools in selecting waste management technologies | | |
| | **OR** | | |
| (c) | Explore the concept of bioresources generation in waste management and give relevant examples. | | |
| **Q4** | (20 marks) | | **CO4** |
| (a) | Categorize biomedical waste and apply waste management knowledge for managing the biomedical waste management? | | |

| | | |
|---|---|---|
| | **OR** | |
| (b) | Analyze how waste management strategies can contribute to reduced environmental footprint and support circular economy. | |
| | **OR** | |
| (c) | Develop a comparison table of thermochemical and biological waste treatment technologies. | |
| **Q5** | (20 marks) | **CO5** |
| (a) | Create a schematic framework explaining how organic waste can be upcycled into different useful products. | |
| | **OR** | |
| (b) | Critically examine how waste materials can be utilized to generate biofuel and bioenergy? | |
| | **OR** | |
| (c) | Give an overview of advances in waste recycling and delivery technologies to deliver value added products | |